Krishna Murthy JATAVALLABHULA

Assistant Professor | Johns Hopkins University

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Berkeley, CA USA

Education

2018-2022	**PhD. in Computer Science**, Université de Montréal, Montréal, Canada. Thesis (letter) grade: **exceptional**.	**GPA: 4.15/4.00**
2015-2017	**MS by research in Computer Science and Engineering**, *International Institute of Information Technology, Hyderabad, India*.	**GPA: 10.00/10.00**
2011-2015	**M.Sc. (Tech.) Information Systems** (Bachelor's degree), *Birla Institute of Science and Technology (BITS), Pilani, India.*	**GPA: 6.71/10.00**

Work

July 2026 – Present | **Assistant Professor, Johns Hopkins University, Computer Science**
Robotics, computer vision, artificial intelligence
Robotics

February 2026 – June 2026 | **Visiting Scholar, Robotics, Toyota Research Institute**
World models, Large behavior models, Post-training, Continual learning, Dextrous manipulation
Robot learning

September 2024 – December 2025 | **AI Research Scientist, FAIR, Meta**
FAIR (Fundamental AI Research) Robotics
Robot learning

March 2022 – September 2024 | **Postdoctoral associate | MIT, (CoCoSci and CSAIL), Cambridge, USA**
With Josh Tenenbaum and Antonio Torralba
Multisensory and multimodal perception | Differentiable probabilistic programming | Physical understanding | Robotics

September 2021 – December 2021 | **Course instructor | McGill University, Montreal, Canada,**
Co-designed and taught *Advanced Image Synthesis* (ECSE 446/546)
Computer graphics | Rendering | Differentiable programming

May 2021 – August 2021 | **Research intern | NVIDIA, Seattle Robotics Group, (Remote)**
With Dieter Fox, Animesh Garg, and Fabio Ramos.
Robotics | Deep learning | Computer graphics | Computer vision

May 2019 – August 2019 | **Deep Learning Research Intern | NVIDIA, Toronto AI Lab, Canada**
With Sanja Fidler. Led the development of Kaolin, a 3D deep learning library for PyTorch.
Deep learning | Computer vision | Computer graphics

Select Honors and Awards

2024 **Microsoft future leader in robotics and AI** Cohort of 14 leading PhD students and postdocs in the US.
2021 **NVIDIA graduate fellowship** One of *5* fellowships awarded worldwide
2021 **Google PhD fellowship** One of *3* fellowships awarded in North America in the *Machine perception, Speech technology, and Computer vision* category (*10* worldwide) (**declined**)
2020 **RSS pioneer 2020**. Selected to the *Robotics Science and Systems pioneers* cohort of 2020, a group of 22 leading senior PhD students and postdocs worldwide.
2020 **Best paper award**. Our paper titled *Maplite: Autonomous intersection navigation without a detailed prior map* won the best paper award for 2020, announced by *Robotics and Automation Letters*.
2021 **Outstanding reviewer** for the IEEE Robotics and Automation Letters, 2020.
2021 **Outstanding reviewer** for the International Conference on Learning Representations
2021 **Outstanding reviewer** for the IEEE international conference on Computer Vision and Pattern Recognition
2020 **Top reviewer** for the *European Conference on Computer Vision* (ECCV), 2020 (1 out of 215 awards)
2019 **DIRO Excellence Award** for research and academic (second consecutive year)

2018 **ICRA PhD Forum**. Selected to present my work at the PhD Forum, ICRA 2018, in my first semester as a PhD student. Received generous travel support.

2018 **DIRO Excellence Award** for research and academic excellence from DIRO, Université de Montréal.

Select Grant Proposals

2026 **Dexmate Inc.** "*Dynamic and dexterous manipulation by autonomous learning from multisensory data*". Co-written with Homanga Bharadhwaj, Anand Bhattad, and Greg Hager

2023 **Army Research Lab**. "*Open-world, Interpretable, Multimodal Models for Intelligent Autonomy*". Co-written with Antonio Torralba and Sarah Schwettmann.

2022 **Army Research Lab**. "*Multimodal generative world models*". Co-written with Antonio Torralba and Josh Tenenbaum.

2020 **IVADO fundamental research grant**. "*Differentiable perception, graphics, and optimization for weakly supervised 3D perception*". Co-written with 3 principal investigators (PI): Liam Paull, James Forbes, Derek Nowrouzezahrai.

2021 **Facebook - unrestricted research gift**. "*Bridging Bayesian optimization and differentiable simulation*". Co-written with Jeannette Bohg (PI) and Rika Antonova (co-PI).

2014 **L K Maheshwari Grant**. Awarded a seed grant for a proposal involving cooperative navigation of a heterogeneous swarm of aerial and ground robots.

Featured publications and preprints

* *equal first-authorship* † indicates *equal advising*

F5. **DREAMSTEER: Latent World Models Can Steer VLA Policies During Deployment Without Any Finetuning**. arXiv 2026
Hanchen Cui, Sergio Arnaud, Arjun Majumdar, Daniel Dugas, Elie Aljalbout, Karthik Desingh†, Krishna Murthy Jatavallabhula†, Franziska Meier†. Project page

F4. **ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning**. ICRA 2024
Qiao Gu*, Ali Kuwajerwala*, Sacha Morin*, **Krishna Murthy Jatavallabhula***, Bipasha Sen, Aditya Agarwal, Corban Rivera, William Paul, Kirsty Ellis, Rama Chellappa, Chuang Gan, Celso Miguel de Melo, Joshua B. Tenenbaum, Antonio Torralba, Florian Shkurti, Liam Paull. Project Page

F3. **ConceptFusion: Open-set Multimodal 3D Mapping**. RSS 2023
Krishna Murthy Jatavallabhula, Alihusein Kuwajerwala, Qiao Gu, Mohd Omama, Tao Chen, Shuang Li, Ganesh Iyer, Soroush Saryazdi, Nikhil Keetha, Ayush Tewari, Joshua B. Tenenbaum, Celso Miguel de Melo, Madhava Krishna, Liam Paull, Florian Shkurti, Antonio Torralba. Project page

F2. **GRADSIM: Differentiable simulation for system identification and visuomotor control**. ICLR 2021
Krishna Murthy Jatavallabhula*, Miles Macklin*, Florian Golemo, Vikram Voleti, Linda Petrini, Martin Weiss, Breandan Considine, Jérôme Parent-Lévesque, Kevin Xie, Kenny Erleben, Liam Paull, Florian Shkurti, Derek Nowrouzezahrai. Video OpenReview

F1. **GRADSLAM: Dense SLAM meets automatic differentiation**. ICRA 2020
Krishna Murthy Jatavallabhula, Ganesh Iyer, Liam Paull. Video Project page

Refereed conference publications

* *equal first-authorship* † *equal advising*

C29. **Locate 3D: Real-World Object Localization via Self-Supervised Learning in 3D**. ICML 2025
Sergio Arnaud*, Paul McVay*, Ada Martin, Arjun Majumdar, **Krishna Murthy Jatavallabhula**, Phillip Thomas, Ruslan Partsey, Daniel Dugas, Abha Gejji, Alexander Sax, Vincent-Pierre Berges, Mikael Henaff, Ayush Jain, Ang Cao, Ishita Prasad, Mrinal Kalakrishnan, Michael Rabbat, Nicolas Ballas, Mido Assran, Oleksandr Maksymets, Aravind Rajeswaran†, Franziska Meier†.

C28. **Gaussian Splatting Visual MPC for Granular Media Manipulation**. ICRA 2025
Wei-Cheng Tseng, Ellina Zhang, **Krishna Murthy Jatavallabhula**, Florian Shkurti.

C27. **ConceptAgent: LLM-Driven Precondition Grounding and Tree Search for Robust Task Planning and Execution**. ICRA 2025
Corban Rivera, Grayson Byrd, William Paul, Tyler Feldman, Meghan Booker, Emma Holmes, David Handelman, Bethany Kemp,

Andrew Badger, Aurora Schmidt, **Krishna Murthy Jatavallabhula**, Celso M de Melo, Lalithkumar Seenivasan, Mathias Unberath, Rama Chellappa.

C26. SparseLoc: Sparse Open-Set Landmark-based Global Localization for Autonomous Navigation. IROS 2025
Pranjal Paul, Vineeth Bhat, Tejas Salian, Mohammad Omama, **Krishna Murthy Jatavallabhula**, Naveen Arulselvan, Madhava Krishna.

C25. SplaTAM: Splat, Track, and Map 3D Gaussians for Dense RGB-D SLAM. CVPR 2024
Nikhil Varma Keetha, Jay Karhade, **Krishna Murthy Jatavallabhula**, Gengshan Yang, Sebastian Scherer, Deva Ramanan, Jonathon Luiten.

C24. PickScan: Object discovery and reconstruction from handheld interactions. IROS 2024
Vincent van der Brugge, Marc Pollefeys, Joshua B Tenenbaum, **Krishna Murthy Jatavallabhula**, Ayush Tewari.

C23. iML: Efficient 3D Instance Mapping and Localization. ICRA 2024
George Tang, Antonio Torralba, **Krishna Murthy Jatavallabhula**.

C22. Talk2BEV: Language-enhanced Bird's-eye View Maps for Autonomous Driving. ICRA 2024
Vikrant Dewangan*, Tushar Choudhary*, Shivam Chandhok*, Shubham Priyadarshan, Anushka Jain, Arun Singh, Siddharth Srivastava, **Krishna Murthy Jatavallabhula**†, Madhava Krishna†.

C21. Anticipate & Act : Integrating LLMs and Classical Planning for Efficient Task Execution in Household Environments. ICRA 2024
Raghav Arora, Shivam Singh, Karthik Swaminathan, Ahana Datta, Snehasis Banerjee, Brojeshwar Bhowmick, **Krishna Murthy Jatavallabhula**, Mohan Sridharan, Madhava Krishna.

C20. Tactile Estimation of Extrinsic Contact Patch for Stable Placement. ICRA 2024
Kei Ota, Devesh K. Jha, **Krishna Murthy Jatavallabhula**, Asako Kanezaki, Joshua B. Tenenbaum.

C19. Learning Correspondence Uncertainty via Differentiable Nonlinear Least Squares. CVPR 2023
Dominik Muhle, Lukas Koestler, **Krishna Murthy Jatavallabhula**, Daniel Cremers.

C18. PAC-NeRF: Physics-augmented continuum neural radiance fields for geometry-agnostic system identification (Spotlight - Top 25% of accepted papers). ICLR 2023
Xuan Li, Yi-Ling Qiao, Peter Yichen Chen, **Krishna Murthy Jatavallabhula**, Ming Lin, Chenfanfu Jiang, Chuang Gan.

C17. Bayesian Object Models for Robotic Interaction with Differentiable Probabilistic Programming. CoRL 2022
Krishna Murthy Jatavallabhula, Miles Macklin, Dieter Fox, Animesh Garg, Fabio Ramos.

C16. Rethinking Optimization with Differentiable Simulation from a Global Perspective (Oral - Top 6.5%). CoRL 2022
Rika Antonova*, Jingyun Yang*, **Krishna Murthy Jatavallabhula**, Jeannette Bohg.

C15. f-Cal: Variational calibration of aleatoric uncertainty in regression. ICRA 2022
Dhaivat Bhatt, Kaustubh Mani, Dishank Bansal, Hanju Lee, **Krishna Murthy Jatavallabhula**, Liam Paull.

C14. Taskography: Evaluating robot task planning over large 3D scene graphs. CoRL 2021
Christopher Agia*, **Krishna Murthy Jatavallabhula***, Mohamed Khodeir, Ondra Miksik, Vibhav Vineet, Mustafa Mukadam, Liam Paull, Florian Shkurti.

C13. DRACO: Weakly Supervised Dense Reconstruction And Canonicalization of Objects. ICRA 2021
Rahul Sajnani, AadilMehdi Sanchawala, **Krishna Murthy Jatavallabhula**, Srinath Sridhar, K. Madhava Krishna.

C12. AutoLay: Benchmarking monocular layout estimation. IROS 2020
Kaustubh Mani, N. Sai Shankar, **Krishna Murthy Jatavallabhula**, K. Madhava Krishna.

C11. Multi-object Monocular SLAM for Dynamic Environments. IV 2020
Gokul Nair, Swapnil Daga, Rahul Sajnani, Anirudha Ramesh, Junaid Ahmed Ansari, **Krishna Murthy Jatavallabhula**, K. Madhava Krishna.

C10. MonoLayout: Amodal scene layout from a single image. WACV 2020
Kaustubh Mani, Swapnil Daga, Shubhika Garg, N. Sai Shankar, **Krishna Murthy Jatavallabhula**, K. Madhava Krishna.

C9. INFER: INtermediate representations for FuturE pRediction. IROS 2019
Shashank Srikanth, Junaid Ahmed Ansari, Karnik Ram R, Sarthak Sharma, **Krishna Murthy Jatavallabhula**, Madhava Krishna.

C8. CalibNet: Geometrically-Supervised Extrinsic Calibration using 3D Spatial Transformer Networks. IROS 2018
Ganesh Iyer, Karnik Ram R., **Krishna Murthy atavallabhula**, K. Madhava Krishna.

C7. **The Earth ain't Flat: Reconstrution of Vehicles on Steep and Bumpy Roads from a Monocular Camera.** IROS 2018
Junaid Ahmed Ansari, Sarthak Sharma, Anshuman Majumdar, **Krishna Murthy Jatavallabhula**, K. Madhava Krishna.

C6. **Constructing Category-Specific Models for Monocular Object SLAM.** ICRA 2018
Parv Parkhiya, Rishabh Khawad, **Krishna Murthy Jatavallabhula**, Brojeshwar Bhowmick, K. Madhava Krishna.

C5. **Beyond Pixels: Leveraging Geometry and Shape Cues for Multi-Object Tracking.** ICRA 2018
Sarthak Sharma, Junaid Ahmed Ansari, **Krishna Murthy Jatavallabhula**, K. Madhava Krishna.

C4. **Shape Priors for Real-Time Monocular Object Localization in Dynamic Environments.** IROS 2017
Krishna Murthy Jatavallabhula, Sarthak Sharma, and K. Madhava Krishna.

C3. **Reconstructing Vehicles From a Single Image: Shape Priors for Road Scene Understanding.** ICRA 2017
Krishna Murthy Jatavallabhula, G.V. Sai Krishna, Falak Chhaya, and K. Madhava Krishna.

C2. **Cluster, Allocate, Cover: An Efficient Approach for Multi-robot Coverage.** SMC 2015
Avinash Gautam, **Krishna Murthy Jatavallabhula**, Gourav Kumar, SP Arjun Ram, Bhargav Jha, and Sudeept Mohan.

C1. **Maxxyt: An Autonomous Wearable Device for Real-Time Tracking of a Wide Range of Exercises.** UKSIM 2015
Danish Pruthi, Ayush Jain, **Krishna Murthy Jatavallabhula**, Ruppesh Nalwaya, and Puneet Teja.

Journal Publications

J7. **STAMP: Differentiable task and motion planning via stein variational gradient descent.** RAL 2025
Yewon Lee, Andrew Z Li, Philip Huang, Eric Heiden, **Krishna Murthy Jatavallabhula**, Fabian Damken, Kevin Smith, Derek Nowrouzezahrai, Fabio Ramos, Florian Shkurti.

J6. **Follow Anything: Open-set detection, tracking, and following in real-time.** RAL 2024
Alaa Maalouf, Ninad Jadhav, **Krishna Murthy Jatavallabhula**, Makram Chahine, Daniel M. Vogt, Robert J. Wood, Antonio Torralba, Daniela Rus.

J5. **Differentiable Visual Computing for Inverse Problems and Machine Learning.** Nature Machine Intelligence 2023
Andrew Spielberg, Cengiz Oztireli, Derek Nowrouzezahrai, Fangcheng Zhong, Konstantinos Rematas, **Krishna Murthy Jatavallabhula**, Tzu-Mao Li.

J4. **AnyLoc: Towards Universal Visual Place Recognition.** RAL 2023
Nikhil Keetha*, Avneesh Mishra*, Jay Karhade*, **Krishna Murthy Jatavallabhula**, Sebastian Scherer, Madhava Krishna, Sourav Garg. Project page

J3. **MapLite: Autonomous intersection navigation without a detailed prior map (Best paper award).** RAL 2020
Teddy Ort, **Krishna Murthy Jatavallabhula**, Rohan Banerjee, Sai Krishna Gottipati, Dhaivat Bhatt, Igor Gilitschenski, Liam Paull, Daniela Rus. Video Paper

J2. **Deep Active Localization.** RAL 2019
Sai Krishna*, Keehong Seo*, Dhaivat Bhatt, Vincent Mai, **Krishna Murthy Jatavallabhula**, Liam Paull.

J1. **FAST: Synchronous Frontier Allocation for Scalable Online Multi-Robot Terrain Coverage.** JIRS 2017
Avinash Gautam, Bhargav Jha, Gourav Kumar, **Krishna Murthy Jatavallabhula**, SP Arjun Ram, and Sudeept Mohan.

Preprints

P2. **ALT-Pilot: Autonomous navigation with Language augmented Topometric maps.** arXiv 2023
Mohammad Omama, Pranav Inani*, Pranjal Paul*, Sarat Chandra Yellapragada, **Krishna Murthy Jatavallabhula**†, Sandeep Chinchali†, Madhava Krishna†.

P1. **Kaolin: A PyTorch Library for Accelerating 3D Deep Learning Research.** arXiv 2019
Krishna Murthy Jatavallabhula, Edward Smith, Jean-Francois Lafleche, Clement Fuji Tsang, Artem Rozantsev, Wenzheng Chen, Tommy Xiang, Rev Lebaredian, Sanja Fidler.

Refereed workshop publications

W7. **ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning.** CoRL Workshops 2023
Qiao Gu*, Ali Kuwajerwala*, Sacha Morin*, **Krishna Murthy Jatavallabhula***, Bipasha Sen, Aditya Agarwal, Corban Rivera, William Paul, Kirsty Ellis, Rama Chellappa, Chuang Gan, Celso Miguel de Melo, Joshua B. Tenenbaum, Antonio Torralba, Florian Shkurti, Liam Paull. Project Page

W6. **ConceptFusion: Open-set Multimodal 3D Mapping.** ICRA Workshops 2023
Krishna Murthy Jatavallabhula, Alihusein Kuwajerwala, Qiao Gu, Mohd Omama, Tao Chen, Shuang Li, Ganesh Iyer, Soroush Saryazdi, Nikhil Keetha, Ayush Tewari, Joshua B. Tenenbaum, Celso Miguel de Melo, Madhava Krishna, Liam Paull, Florian Shkurti, Antonio Torralba. Project page

W5. **RobustPointSet: A Dataset for Benchmarking Robustness of Point Cloud Classifiers.** ICLR Workshops 2021
Saeid Asgari Taghanaki, Jieliang Luo, Ran Zhang, Ye Wang, Pradeep Kumar Jayaraman, **Krishna Murthy Jatavallabhula**.

W4. **GradSLAM: Automagically differentiable SLAM.** CVPR Workshops 2020, RSS Workshops 2020
Krishna Murthy Jatavallabhula, Ganesh Iyer, Soroush Saryazdi, Liam Paull. Video Project page

W3. **Probabilistic object detection: Strengths, Weaknesses, Opportunities.** ICML Workshops 2020
Dhaivat Bhatt, Dishank Bansal, Gunshi Gupta, Hanju Lee, **Krishna Murthy Jatavallabhula**, Liam Paull.

W2. **Reconstruct, Rasterize and Backprop: Dense shape and pose estimation from a single image.** CVPR Workshops 2020
Aniket Pokale, Aditya Aggarwal **Krishna Murthy Jatavallabhula**, K. Madhava Krishna.

W1. **Geometric Consistency for Self-Supervised End-to-End Visual Odometry.** CVPR Workshops 2018
Ganesh Iyer*, **Krishna Murthy Jatavallabhula***, Gunshi Gupta, K. Madhava Krishna, and Liam Paull..

Professional Service and Volunteering

2017-Present	Reviewer; robotics (ICRA, IROS, RAL, RSS, CoRL), Vision (CVPR, ICCV, ECCV, ACCV, WACV, ICVGIP, CRV), and ML (Neurips, ICML, ICLR, AAAI) venues
2022-present	Associate editor; IROS
2025	Area chair; CVPR, CoRL
2024	OpenReview chair; Conference on Robot Learning (CoRL)
2024	Associate editor; IEEE Robotics and Automation Letters
2023	Publicity and social media chair; Canadian AI Conference
2020-2021	Student Volunteer, ICML (International Conference on Machine Learning)
2020	Student Volunteer, RSS (Robotics Science and Systems)
2020-2021	Student Volunteer, ICLR (International Conference on Learning Representations)

Outreach and Inclusion

2022-2024	Mentor - Mila mentorship program
2022-2024	Mentor - Black in AI academic program
2021	Student member, Mila equity, diversity, and inclusion (EDI) committee (1 of 7 student representatives)
2020	Mentor, Neurips workshop (DiffCVGP)
2020	Diversity and inclusion panel, RSS (Robotics Science and Systems)
2018	Mentor, AI for social good workshop. McGill University.

Workshops and Sessions Co-organized

Jun 2024	Organizer, Multimodalities for 3D scenes (CVPR 2024 workshop)
Dec 2021	*Lead organizer* (Proposer, Program chair), Physical reasoning and inductive biases for the real world (Neurips 2021 workshop) Webpage
Oct 2021	*Lead organizer* (Proposer, Program chair), Differentiable 3D computer vision and graphics (ICCV 2021 workshop). Webpage
Jul 2021	*Program co-chair*, Robotics Science and systems pioneers workshop (RSS 2021). Webpage
May 2021	*Lead organizer* (Proposer, Program chair), Beyond the research paper: Rethinking how we share scientific understanding in ML (ICLR 2021 workshop). Webpage
Jan-May 2021	*Lead Organizer*, Robot learning seminar series: Mila and REAL - Winter 2021. Webpage
Dec 2020	*Lead organizer* (Proposer, Program chair), Differentiable vision, graphics, and physics applied to machine learning (Neurips 2020). Webpage
Sep-Dec 2020	*Lead Organizer*, Robot learning seminar series: Mila and REAL - Fall 2020. Webpage
Nov 2019	*Breakout session organizer*, Pan-Canadian SOCMLx.

Talks

June 25 2025	Invited talk - Workshop on learned robot representations, RSS 2025
June 20 2025	Invited talk - Unifying visual SLAM worksthop, RSS 2025
May 23 2025	Invited talk - Neuro AI group meeting, University of Washington
Jun 17 2024	Invited keynote - CVPR 2024 Workshop on Scene Graphs and Graph Representation Learning
Jun 7 2024	Invited talk - Mohamed Bin Zayed University of Artificial Intelligence
Apr 26 2024	Group meeting speaker - SPARK Lab, MIT
Apr 12 2024	Job talk - CS and ECE departments at Johns Hopkins University
Mar 13 2024	Job talk - Meta AI (FAIR)
Mar 1 2024	Invited talk - Microsoft/UMD future leaders in robotics and AI
Feb 27 2024	Invited talk - School of Interactive Computing, Georgia Institute of Technology
Feb 21 2024	Group meeting speaker - Toronto Intelligent Systems Lab - University of Toronto
Feb 20 2024	Group meeting speaker - Robot vision and learning lab - University of Toronto
Nov 17 2023	Guest lecture - Robot Learning Course - University of Illinois Urbana-Champaign
Oct 31 2023	Invited talk - University of Bristol - ML and CV seminar (MaVi)
Oct 17 2023	Invited talk - Boston Dynamics
June 6 2023	Guest lecture - Computer vision course offering
Apr 24 2023	Invited talk - Stanford SVL
Apr 24 2023	Invited talk - Microsoft Mixed Reality Seminars
Mar 23 2023	Invited talk at Cornell Robotics
Feb 23 2023	Invited talk - Katerina Fragkiadaki's group (CMU)
Feb 22 2023	Invited talk - Scene representations group (MIT)
Jan 14 2023	Invited talk - IIT Kanpur robotics club
Dec 16 2022	Guest lecture - ROBGY 6203 - Robot perception - New York University (NYU)
Dec 2 2022	MIT BCS physical reasoning meeting - **Physical understanding: An AI perspective**
Dec 1 2022	Guest speaker - MIT EECS 6.S980 - Machine learning for inverse graphics (Instructor: Vincent Sitzmann)
Oct 6 2022	Invited talk - MIT Machine Intelligence Research Forum - **Differentiable programming for spatial AI**
Dec 2021	Invited talk - Talking robotics series
Nov 2021	Guest lecture - Introduction to autonomous vehicles (Duckietown) - Université de Montréal
Oct 2021	Structural and Compositional Learning on 3D Data, ICCV 2021 Workshop - **Taskography: Task planning over large 3D scene graphs**
Aug 2021	AI for Autonomous Driving workshop, IJCAI 2021
July 2021	Tartan SLAM series - Carnegie Mellon University
June 23 2021	Invited talk - ML reading group at the University of Sydney
June 15 2021	Invited talk - Dynamical systems reading group, Mila
Apr 7 2021	Microsoft autonomous systems - **gradSim: A differentiable simulation framework**
Mar 26 2021	AI in robotics (University of Toronto) - **gradSLAM + gradSIM**
Feb 23 2021	KUIS AI (Istanbul) - **Building differentiable models of the 3D world**
Jan 19 2021	MIT Vision seminar - **Building differentiable models of the 3D world**
Oct 11 2020	IEEE chapter, Indonesia - **Deep learning for robot perception**
Sep 22 2020	Cornell robotics group - **gradSLAM: Dense SLAM meets automatic differentiation**
Aug 29 2020	CV Talks, India: Computer vision talks - **gradSLAM: Automagically differentiable SLAM**
Jul 2020	Robotics Science and Systems pioneers - **gradSLAM: Dense SLAM meets automatic differentiation**
Jul 2020	Robotics Science and Systems: structured approaches to robot learning workshop - **gradSLAM: Automagically differentiable SLAM**
Jun 2020	CVPR: Deep declarative networks workshop - **gradSLAM: Automagically differentiable SLAM**
Feb 2019	NVIDIA Webinar - **3D deep learning with Kaolin**

Teaching

2021	(Instructor) **Realistic / Advanced image synthesis** (ECSE 446/546) at McGill, Montreal.
2021	(Teaching assistant) **Representation Learning** at Mila and Université de Montréal, with Aaron Courville.
2020	(Teaching assistant) **Advanced projects in deep learning** at Mila, with Pierre-Luc Carrier and Joumana Ghosn.
2017	(Designed and co-taught) **Mobile Robotics and Computer Vision** at IIIT Hyderabad, with Prof. K. Madhava Krishna.
2016	(Teaching assistant) **Mobile Robotics** at IIIT Hyderabad, with Prof. K. Madhava Krishna.

Students Mentored

A list of students I have closely mentored on a research or technical project. (Criteria: Mentorship lasted 3 months or longer)

7 Students at their PhD level or equivalent.
30 Students pursuing Masters programs
20 Students at their undergraduate level of study (includes UROP students at MIT, visitors / interns at Mila, Université de Montréal and IIIT Hyderabad, India)